

A smart mask that measures city pollution

It is common these days to see people wearing masks to keep out dust, other pollutants and airborne infections. This is so in China too. Due to the poor air quality there, current-generation Chinese people are finicky about preventive healthcare, and therefore there are a lot of bulletins, mobile apps, etc that give updates about air quality there, quite like traffic updates. However, there is another catch there, as the Chinese are also very sceptical of faceless data. Hence Frog Design has come up with an air mask design called AirWaves that serves a dual purpose: it acts as an air mask, and also helps read and feed air quality data into a central system. By leveraging community and bringing granularity into air quality data, it is expected to make the information more trustworthy. However, in an interview with BBC, one of the company's directors admitted that there will be several political and governmental discussions to get through before this product is brought to the market, as the air quality is a key aspect of socio-economic development and is a 'loaded' topic.

What's inside: By adding particle sensors to an air mask, AirWaves combines the filtering and measurement of air quality into a single product. The Bluetooth-enabled mask reads many parameters of air quality, and passes on this data to an associated mobile app, which feeds it into a larger network. The app also lets you browse the consolidated, real-time data at the desired level of detail: in your city, neighbourhood or places you will be passing on your route. The project is still in an early stage and many challenges have to be overcome before it can be even prototyped. For example, there is the mechanical challenge of how to fit a particle filter, model, battery and power management unit into the mask without compromising flexibility and comfort. It is likely that the end-product will be a combination of a plastic shell to house the components, and a softer material at the skin contact points, not to forget the traditional filter tissue. All of these might get done soon, since the project is in the hands of Azure Yang and Mingmin Wang—two of Frog's best designers. Once the product is designed, they will start working on the connectivity and software aspects.

Web: http://info.frogdesign.com/rs/aricent/images/frog_Mobile_Ecosystems-Wearable-Technology.pdf



A shape-changing smartphone

We tend to put our phones on silent when we get into a meeting, but sometimes the vibration is so subtle that we do not notice an incoming call unless we are holding the phone at that time. In an attempt to overcome this problem, researchers at the Ontario-based Queen's University have developed the MorePhone—an instrument that curls up when a call arrives! Developed using thin, flexible displays, the smartphone can morph its shape to give users a silent yet visual cue of an incoming phone call, text message or email. The MorePhone has been developed by Dr Roel

Vertegaal, director of the Human Media Lab, School of Computing, Queen's University, and his students. He thinks bendable, flexible cell phones are the future and MorePhone could be in the market within five to ten years.

What's inside: MorePhone is made of a thin, flexible electrophoretic display manufactured by Plastic Logic—a British company and a world leader in plastic electronics. Sandwiched beneath the display are a number of shape-memory alloy (SMA) wires that contract during a notification, causing the phone to curl either its entire body or

up to three individual corners. Users of the MorePhone can customise how they want each notification to be. For example, they can set one corner to curl up when an SMS arrives, two corners to curl up when an email arrives, and three corners to curl up when a call arrives—or any other way they are comfortable with. They can also configure the corners to keep curling repeatedly when a high-priority message or call arrives. This curling of the MorePhone is made possible by the use of SMA—an alloy that remembers its original, cold-forged shape. After curling, it can return to the original shape when heated. SMAs are considered a light-weight, solid-state alternative to motors and actuators, and generally used in medical and aerospace applications. It is interesting to note its use in phones.

Web: <http://www.hml.queensu.ca/blog/morephone>

